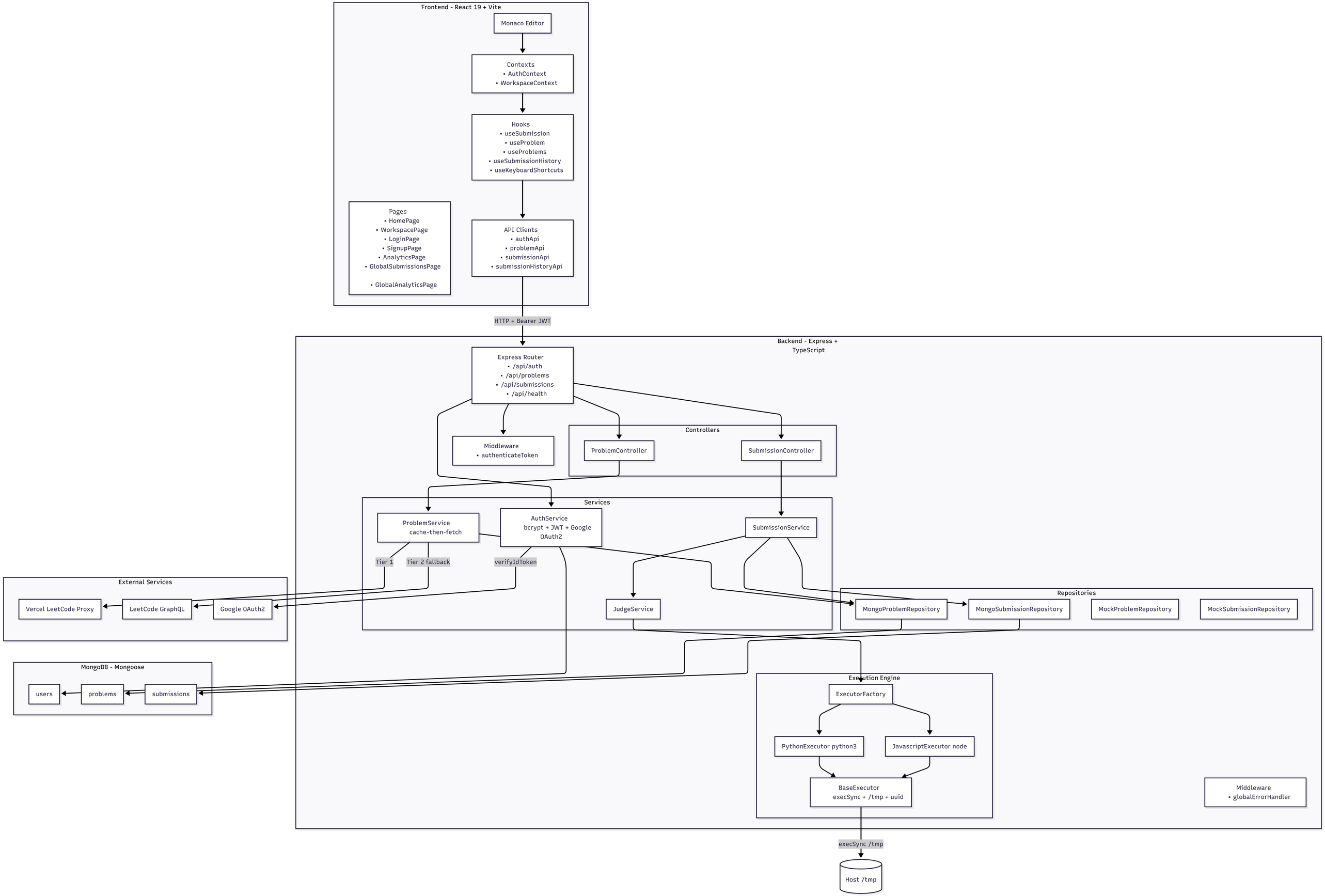
Frontend - React 19 + Vite
Monaco Editor
Contexts
• AuthContext
• WorkspaceContext
Hooks
• useSubmission
• useProblem
• useProblems
• useSubmissionHistory
• useKeyboardShortcuts
Pages
• HomePage
• WorkspacePage
• LoginPage
• SignupPage
• AnalyticsPage
• GlobalSubmissionsPage
• GlobalAnalyticsPage
API Clients
• authApi
• problemApi
• submissionApi
• submissionHistoryApi
HTTP + Bearer JWT
Backend - Express + TypeScript
Express Router
• /api/auth
• /api/problems
• /api/submissions
• /api/health
Middleware
• authenticateToken
Controllers
ProblemController
SubmissionController
Services
ProblemService
cache-then-fetch
AuthService
bcrypt + JWT + Google OAuth2
SubmissionService
Tier 1
Tier 2 fallback
verifyIdToken
JudgeService
Repositories
MongoProblemRepository
MongoSubmissionRepository
MockProblemRepository
MockSubmissionRepository
External Services
Vercel LeetCode Proxy
LeetCode GraphQL
Google OAuth2
MongoDB - Mongoose
users
problems
submissions
Execution Engine
ExecutorFactory
PythonExecutor python3
JavascriptExecutor node
BaseExecutor
execSync + /tmp + uuid
Middleware
• globalErrorHandler
execSync /tmp
Host /tmp